

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GLUE | Context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotator demographics for the main inherited GLUE tasks are not documented in the paper [Q6 establishes the inherited-dataset reliance; no demographic disclosure for SST-2 [Q26] or MNLI crowdworkers [Q36]]. The diagnostic set's annotators are explicitly six NLP researchers [Q73] with high inter-annotator agreement (Fleiss's $\kappa = 0.73$ [Q74], $R3 = 0.80$ [Q75]) — but this is a specialized English-speaking population with no documented connection to Luxembourgish administrative culture or continental-European formal correspondence register. Ground-truth sentiment in SST-2 reflects English movie-review judgments (DATASET-D1 through D5, D29, D30); ground-truth entailment in MNLI/RTE reflects English crowdsourced judgments. The dataset analysis shows SST-2 fragments often labelled in ways that depend entirely on film-criticism context [DATASET-D2, D29, D30]. Luxembourg's understated formal administrative register, where frustration is expressed indirectly (cultural_norms_notes), is not represented in any annotator pool. The user explicitly flagged OC as a 'live risk' (elicitation A3). Even ItzGLUE [WEB-10] and the only Luxembourgish sentiment corpus [WEB-25] do not cover administrative register — the latter sources from RTL.lu informal news comments.

ID	Evidence
Q6	"None of the datasets in GLUE were created from scratch for the benchmark; we rely on preexisting datasets because they have been implicitly agreed upon by the NLP community as challenging and interesting." (p.1)
Q26	"The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment." (p.3)
Q36	"The Multi-Genre Natural Language Inference Corpus (Williams et al., 2018) is a crowdsourced collection of sentence pairs with textual entailment annotations." (p.4)
Q73	"To establish human baseline performance on the diagnostic set, we have six NLP researchers annotate 50 sentence pairs (100 entailment examples) randomly sampled from the diagnostic set." (p.6)
Q74	"Inter-annotator agreement is high, with a Fleiss's κ of 0.73." (p.6)
Q75	"The average R3 score among the annotators is 0.80, much higher than any of the baseline systems described in Section 5." (p.6)
D1	sleepwalk through vulgarities in a sequel you can refuse — English movie-review fragment; sentiment label calibrated to film criticism register
D2	at its best moments — Sub-sentential movie-review fragment; context-free without film domain
D5	is a pan-american movie , with moments of genuine insight into the urban heart . — Movie-review sentence referencing pan-American themes
D29	the horrors — Two-word fragment; highly context-dependent sentiment label
D30	heroes — Single-word fragment; no sentential context
WEB-10	ItzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)
WEB-25	RTL Corpus Sentiment Annotation Tool (STRIPS) — only known Luxembourgish sentiment resource, sourced from informal RTL news comments rather than administrative register (https://aclanthology.org/2020.sltu-1.22.pdf)

Strengths

- GLUE reports inter-annotator agreement (Fleiss's $\kappa = 0.73$ [Q74]) and a human baseline ($R3 = 0.80$ [Q75]) for the diagnostic set, establishing a methodological precedent for human-baseline reporting that any custom Luxembourgish

benchmark should follow.

- The diagnostic set was tested for hypothesis-only artifacts using fastText classifiers (near-chance 32.7% / 36.4% [Q72]), demonstrating attention to label-quality verification — a transferable practice.

ID	Evidence
Q72	"The models respectively get near-chance accuracies of 32.7% and 36.4% on our diagnostic data, showing that the data does not suffer from such artifacts." (p.6)
Q74	"Inter-annotator agreement is high, with a Fleiss's κ of 0.73." (p.6)
Q75	"The average R3 score among the annotators is 0.80, much higher than any of the baseline systems described in Section 5." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground-truth labels reflect English-speaking annotator pools (movie reviewers, crowdworkers, NLP researchers [Q26, Q36, Q73]) with no documented Luxembourgish or continental-European administrative-culture representation.

OC-2: Significant disagreement is expected. Luxembourg administrative correspondence uses understated formal register (cultural_norms_notes); frustration is rarely expressed directly. English movie-review sentiment annotators [DATASET-D1, D3, D29, D30] are calibrated to overt evaluative language. A label of 'positive sentiment' on a fragment like 'at its best moments' [DATASET-D2] has no analogue in formal Luxembourgish administrative messages.

OC-3: Annotator demographics are NOT documented for the main benchmark tasks. The paper does not describe SST-2 sentiment annotators [Q26] or MNLI crowdworker demographics [Q36], nor inter-annotator agreement for inherited datasets [Q6]. Only the diagnostic set's annotators are described — six NLP researchers [Q73].

OC-4: Re-annotation by Luxembourgish civil servants would be required. The deployment's human-in-the-loop mechanism (elicitation A3) is positioned to produce such corrections post-hoc, but baseline labels in any GLUE-style training/evaluation would still encode the original misalignment.

OC-5: GLUE's aggregation methods are not documented at the per-example level for inherited datasets. The diagnostic set uses majority labelling with high κ [Q74] but only six annotators from a homogeneous population — minority perspectives could be erased without detection. Multi-label phenomena tagging [Q141] is permitted but does not extend to label disagreement modeling.

OC-6: Convergent-validity violation is severe: labels do not correlate with judgments of regional stakeholders in administrative culture. External-validity violation is also severe: judgments of English film reviewers and US crowdworkers will not generalize to Luxembourg civil servants assessing citizen correspondence.

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Q26	"The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment." (p.3)

Q36	“The Multi-Genre Natural Language Inference Corpus (Williams et al., 2018) is a crowdsourced collection of sentence pairs with textual entailment annotations.” (p.4)
Q73	“To establish human baseline performance on the diagnostic set, we have six NLP researchers annotate 50 sentence pairs (100 entailment examples) randomly sampled from the diagnostic set.” (p.6)
Q74	“Inter-annotator agreement is high, with a Fleiss's κ of 0.73.” (p.6)
Q141	“Note that some examples may be tagged with phenomena belonging to multiple categories.” (p.15)
D1	sleepwalk through vulgarities in a sequel you can refuse — English movie-review fragment; sentiment label calibrated to film criticism register
D2	at its best moments — Sub-sentential movie-review fragment; context-free without film domain
D3	the weird thing about the santa clause 2 , purportedly a children 's movie , is that there is nothing in it to engage children emotionally . — Full sentence from American movie review; US pop-culture reference
D29	the horrors — Two-word fragment; highly context-dependent sentiment label
D30	heroes — Single-word fragment; no sentential context
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Information Gaps

- Demographic composition of MNLI crowdworkers and SST-2 sentiment annotators is not disclosed in the paper.
- Inter-annotator agreement metrics for inherited GLUE component datasets are not reported in the GLUE paper [Q6 implies inheritance without re-reporting].

Requires Expert Verification

- A pilot re-annotation of administrative-correspondence sentiment by Luxembourgish civil servants would establish empirical disagreement rates and quantify the OC gap.

Remediation

Gap 1: Ground-truth labels in GLUE reflect English-speaking annotator pools with no connection to Luxembourgish administrative register; SST-2 sentiment is calibrated to film criticism, not understated formal correspondence.

Recommendation: Recruit Luxembourgish civil servants familiar with administrative correspondence for a custom annotation task, using clear annotation guidelines for understated frustration/urgency signals. Report inter-annotator agreement following GLUE's diagnostic-set methodology [Q73, Q74] but with a representative annotator pool. Use the deployment's human-in-the-loop corrections (elicitation A3) as an ongoing label-validation feedback channel.

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Q73	“To establish human baseline performance on the diagnostic set, we have six NLP researchers annotate 50 sentence pairs (100 entailment examples) randomly sampled from the diagnostic set.” (p.6)
Q74	“Inter-annotator agreement is high, with a Fleiss's κ of 0.73.” (p.6)